Summary :

📌 Overview

Using K-Means clustering, songs were grouped based on audio features such as:

Energy
Danceability
Valence
Acousticness
Speechiness
Tempo
The clustering reveals distinct musical behaviors and moods.

🟢 Cluster 0 — Upbeat / Lively

🔍 Characteristics

High danceability → easy to move/dance
Moderate to high energy → lively sound
High valence → positive, happy mood
Moderate tempo → rhythmic and engaging
🎧 Interpretation

These are:

Party songs
Pop / dance tracks
Feel-good music
💡 Insight

👉 This cluster represents mainstream popular music
👉 Likely to dominate playlists and streaming platforms

🔵 Cluster 1 — Narrative / Spoken Word

🔍 Characteristics

High speechiness → more spoken content
Lower musical complexity
Often lower energy

🎧 Interpretation

These include:

Rap / hip-hop (lyric-heavy)
Podcasts / spoken word
Storytelling music
💡 Insight

👉 This cluster is content-driven rather than melody-driven
👉 Shows how speech patterns influence clustering

🟡 Cluster 2 — Ambient / Mellow

🔍 Characteristics

Low energy → soft and calm
Low valence → neutral or slightly emotional
High acousticness → natural instruments
Lower tempo
🎧 Interpretation

These are:

Relaxation music
Instrumental / ambient tracks
Study / focus playlists
💡 Insight

👉 Represents calm listening behavior
👉 Strongly associated with background or passive listening

🟠 (Optional) Cluster 3 — Chill / Acoustic (if present)

🔍 Characteristics

High acousticness
Low to moderate energy
Balanced valence
🎧 Interpretation

Indie acoustic songs
Soft vocals
Coffee-shop style music
📊 Cross-Cluster Insights

🔷 1. Feature Importance

Key features driving clustering:

Energy → intensity
Valence → emotion
Acousticness → sound texture
Speechiness → vocal style
🔷 2. Behavioral Segmentation

The model separates music into:

Type	Cluster
Active listening (party/workout)	Cluster 0
Content-focused listening	Cluster 1
Passive/relax listening	Cluster 2

🔷 3. Music Consumption Insight

👉 Users typically fall into:

High-energy listeners
Content listeners
Relaxation listeners
🔷 4. Business Value

This clustering can be used for:

🎧 Playlist generation
🤖 Recommendation systems
📈 Music trend analysis
🎯 Targeted user segmentation
🎯 Final Conclusion

The clustering successfully identifies distinct music listening patterns, transforming raw audio features into meaningful categories like Upbeat, Narrative, and Ambient.